

Exponential-Range Mass Production of Boolean Functions

Local disjoint recovery and recursive congestion reduction

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August 2026

Abstract

For a Boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}$, let $f^{\times t}$ denote the multi-output function that evaluates f on t independent inputs. A theorem of Uhlig shows that $t = 2^{o(n/\log n)}$ copies can be computed with essentially the worst-case circuit size needed for one copy. We give a different construction that reaches every fixed exponential rate below the 2^n -copy scale: for every fixed $0 \leq \gamma < 1$,

$$C(f^{\times t}) = O_\gamma(2^n/n) \quad \text{for all } t \leq 2^{\gamma n}.$$

This establishes the optimal worst-case order of growth, but does not retain Uhlig’s one-copy leading constant.

The construction encodes restrictions of f by a constant-rate multivariate evaluation code. Each requested restriction is recovered from the non-target points of an affine line. A deterministic greedy scheduler makes these line recoveries disjoint inside request groups. Different groups create bounded congestion, which is absorbed by a recursive call on fewer input variables. At the level of admissible exponents, this gives the bootstrap

$$\beta \mapsto \frac{1}{2 - \beta}.$$

Starting from the disjoint range $\beta = 1/2$, iteration reaches every constant below one.

1 Introduction

For a Boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}$ and an integer $t \geq 1$, define

$$f^{\times t}(x^{(1)}, \dots, x^{(t)}) = (f(x^{(1)}), \dots, f(x^{(t)})).$$

The naive construction uses t disjoint copies of a circuit for f and gives

$$C(f^{\times t}) \leq tC(f).$$

For unrestricted circuits this direct-sum bound can be far from optimal. Uhlig’s mass-production theorem allows $t = 2^{o(n/\log n)}$ independent evaluations with no asymptotic increase in the leading worst-case one-copy synthesis cost; the numerical leading constant depends on the circuit basis and cost convention. See Uhlig’s later account [12], Wegener’s Theorem 10.2.3 [13], and the modern restatement in [6]; the 1974 paper [11] is a precursor.

We prove, by a different construction, an order-of-growth bound for $2^{\gamma n}$ simultaneous evaluations for every fixed $\gamma < 1$. The price is that the proof controls only the optimal worst-case order $2^n/n$, not the sharp one-copy leading constant.

1.1 Main result

We fix the standard basis $\{\wedge, \vee, \neg\}$, with fan-in two for the binary gates. Fan-out and output designations are free, and circuit size is the number of non-input gates. Any other fixed finite complete basis changes the result by at most a constant-factor simulation.

Theorem 1.1 (Exponential-range mass production). *For every fixed $0 \leq \gamma < 1$, there is a constant A_γ such that, for all sufficiently large n , every Boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}$, and every integer $1 \leq t \leq 2^{\gamma n}$,*

$$C(f^{\times t}) \leq A_\gamma \frac{2^n}{n}.$$

The following elementary bound does not explain the fixed-exponent gap, but it shows that the plateau cannot extend to the 2^n -copy scale. If $n \geq 2$ and f depends on all n variables, then every circuit for $f^{\times t}$ has size at least $tn/2$. Consequently, a uniform $O(2^n/n)$ plateau cannot persist once $t = \omega(2^n/n^2)$; see the discussion after [Corollary 8.1](#). Thus [Theorem 1.1](#) reaches every fixed exponential rate while leaving a genuine near-endpoint transition to be understood.

The proof separates an input into a prefix and a suffix. For every fixed suffix, the 2^p prefix restrictions are packed into information symbols of a constant-rate low-degree evaluation code over a field $\mathbb{F}_q$. A requested information symbol can be recovered from the $q - 1$ non-target points on any affine line through it. There are about $q^{\ell-1}$ possible directions.

Within one group of requests, a greedy circuit selects pairwise disjoint line recovery sets. Across C groups, a code coordinate is used at most C times. The repeated evaluations at one coordinate are then performed by a circuit for a C -fold direct product on the shorter suffix input. This is the recursive step: local disjointness reduces global sharing to a smaller instance of the same problem.

1.2 The exponent bootstrap, in outline

Here is the parameter logic behind the recursion. To handle $t \leq 2^{\gamma n}$ requests, the suffix has length $d \approx \gamma n$ and the number C of request groups has exponent c , so a resource sees at most 2^{cn} suffixes. A previously admissible exponent β handles that recursive demand when $c < \beta\gamma$, while the scheduling overhead requires $c > 2\gamma - 1$, up to slack that vanishes as the fixed number ℓ of code variables grows. Thus a feasible c exists precisely when $\gamma < 1/(2 - \beta)$.

Writing $\text{Adm}(\beta)$ for the admissible-exponent property defined formally in [Definition 7.1](#), a globally disjoint schedule gives $\text{Adm}(1/2)$, and the grouped construction proves

$$\text{Adm}(\beta) \implies \text{Adm}\left(\frac{1}{2 - \beta}\right).$$

Starting from $\beta_1 = 1/2$, the resulting sequence is

$$\beta_k = \frac{k}{k + 1},$$

which converges to one.

1.3 A quantitative consequence

Suppose a nonnegative measure μ is bounded above by circuit size and assigns at least t times as much complexity to t independent copies. The theorem gives, for every fixed $\varepsilon > 0$,

$$\mu(f) = O_\varepsilon(2^{\varepsilon n}/n)$$

uniformly over $f : \{0, 1\}^n \rightarrow \{0, 1\}$. Paul’s earlier general many-copy bound [9] already implies the qualitative conclusion that such a measure is uniformly subexponential; our bound improves its polynomial prefactor from $O_\varepsilon(n2^{\varepsilon n})$ to $O_\varepsilon(2^{\varepsilon n}/n)$. Any measure capable of proving an exponential unrestricted-circuit lower bound must therefore account for cross-instance reorganization rather than charging each copy independently.

1.4 Relation to earlier work

The theorem-level comparisons we were able to verify are summarized below. Here “one-copy leading term” refers to the basis-dependent Shannon–Lupanov worst-case asymptotic.

Result	Copy range	Circuit-size bound	Comparison
Uhlig [12, 13, 6]	$\log t = o(n/\log n)$	one-copy leading term	optimal order and leading constant in a smaller copy range
Paul, Lemma 2 [9]	arbitrary t	$O(\max\{n2^n, nt \log^2 t\})$	$O(n2^n)$ throughout every fixed exponential range below one
This paper	$t \leq 2^{\gamma n}$ for fixed $\gamma < 1$	$O_\gamma(2^n/n)$	optimal worst-case order, without leading-constant control

For fixed $\gamma < 1$ and $t \leq 2^{\gamma n}$, Paul’s second term is $o(n2^n)$, so his displayed bound is $O(n2^n)$. The theorem proved here improves that bound by a factor of n^2 at the level of polynomial prefactors. Paul’s g^k is the same k -fold product task considered here; his circuits permit arbitrary binary Boolean gates, which changes our fixed basis cost by at most a constant factor.

Albrecht’s six-page paper studies the same simultaneous-realization problem over a complete binary basis [1]. Its abstract states that, for the corresponding Shannon size and depth functions, a relatively large number of simultaneous inputs is compatible with a small enhancement. The available primary-source preview contains the circuit model and opening definitions but not the quantitative theorem statements. We therefore do not assign Albrecht a copy range or size bound in the table, and we make no historical priority claim or claim that a leading-constant question is open. A definitive comparison requires checking the full paper.

The coding interpretation begins with Uhlig’s two-copy construction, which is the length- $(2^p + 1)$ single-parity-check code viewed as a two-request disjoint recovery code. Systematic multivariate polynomial evaluation, recovery on affine lines, and the use of disjoint recovery sets appear in the batch-code literature, especially Section 3.4 of Ishai–Kushilevitz–Ostrovsky–Sahai [5]. Their geometric procedure chooses lines randomly and deletes intersections. The construction here instead gives a deterministic greedy selection circuit, partitions requests into groups, recursively absorbs the resulting inter-group congestion, and charges the scheduler and fixed-wire routing explicitly. We make no priority claim for these refinements.

A 2011 manuscript of Lipmaa, subsequently withdrawn, proposed computational batch codes in connection with multi-query computation [7]. Later lifted-polynomial constructions also use line-based recovery to obtain batch-code guarantees [4]. PIR codes of Fazeli, Vardy, and Yaakobi form a related family in which each information symbol has many disjoint recovery options [3]. We use only the elementary product-code and affine-line facts proved below.

Independently of historical priority, the technical contribution established here is a self-contained deterministic disjoint-line scheduler, an explicit fixed-wire scatter/gather implementation, and a recursive bounded-congestion invariant whose exponent map is $\beta \mapsto 1/(2 - \beta)$.

2 Circuit conventions and worst-case notation

Our circuits are finite directed acyclic graphs. Source vertices are input bits or the constants zero and one. Internal vertices are labelled by $\wedge$, $\vee$, or $\neg$, with the usual fan-in. A circuit for a map $F : \{0, 1\}^N \rightarrow \{0, 1\}^m$ has m designated output wires, which may be input wires or gate outputs and may repeat. Fan-out, constants, and output designations carry no cost; size is the number of internal gates. We write $C(F)$ for the minimum size of such a circuit. All leading-constant comparisons in this paper refer to this fixed model.

Let

$$L(n) = \max_{f: \{0,1\}^n \rightarrow \{0,1\}} C(f).$$

The Shannon counting lower bound and Lupanov's synthesis upper bound imply

$$L(n) = \Theta(2^n/n)$$

for every fixed finite complete basis; see [10, 8, 13]. We use only the upper bound.

For $r \geq 1$, define

$$L_r(n) = \max_{f: \{0,1\}^n \rightarrow \{0,1\}} C(f^{\times r}).$$

For every $r \geq 1$, one has $L_r(n) \geq L(n)$: fix all but one of the r input blocks and retain the corresponding output. Thus $2^n/n$ is the optimal worst-case order even when several outputs are requested.

For each f , choose a circuit for $f^{\times r}$ of size at most $L_r(n)$. If a construction supplies fewer than r live inputs, the unused inputs of this circuit may be fixed arbitrarily and their outputs ignored. Thus a bound for $L_r(n)$ also applies to every smaller number of copies.

All circuits below are nonuniform. Field representations, irreducible polynomials, sorting networks, and other finite combinatorial objects may be fixed as part of the circuit for each input length.

We use $\tilde{O}_\ell(X)$ for

$$X \cdot \text{poly}_\ell(n, \log q, \log(t + 2), \log(C + 2)).$$

Only polynomial factors are suppressed; in the parameter regimes used below, they never affect an exponential rate.

For a positive integer k , write $[k]_0 = \{0, \dots, k - 1\}$.

We repeatedly use two standard circuit primitives. First, after representing $\mathbb{F}_{2^b}$ in a polynomial basis, field addition, multiplication, inversion of a nonzero element, and comparison of canonical

b -bit representations have circuits of size $\text{poly}(b)$. For example, inversion can be implemented by binary exponentiation to the power $2^b - 2$. Second, R records of width w can be sorted by a fixed sorting network using $O(R \log^2 R)$ compare-exchange units, each of size $O(w)$; see [2]. Sorting by a leading type bit, with original positions used as tie breakers when needed, also gives stable fixed-wire compaction within the same asymptotic bound.

Sharing as bounded simultaneous demand. The organizing principle is that one physical resource can serve equal data-dependent requests, while distinct simultaneous arguments require separate demand slots. Below, a resource coordinate G_y is requested on at most one suffix inside each request group. With C groups, its simultaneous demand is at most C . The circuit pays explicitly for selecting, routing, and decoding those demands, while a C -fold direct-product circuit evaluates the slots jointly. This is how local disjointness becomes global sharing in the main construction.

3 The two-copy construction as disjoint recovery

Fix a prefix length p , put $M = 2^p$, and define the restrictions

$$f_i(z) = f(i, z), \quad i \in \{0, \dots, M-1\}.$$

Uhlig's two-copy construction introduces the following functions; see the proof of Wegener's Theorem 10.2.3 [13] and the overview in [6, Section 1.2]:

$$g_0 = f_0, \quad g_i = f_{i-1} \oplus f_i \quad (1 \leq i < M), \quad g_M = f_{M-1}.$$

Every restriction has two representations:

$$f_i = \bigoplus_{j=0}^i g_j = \bigoplus_{j=i+1}^M g_j.$$

Given requests $i \leq j$, recover f_i from the prefix set $\{0, \dots, i\}$ and f_j from the suffix set $\{j+1, \dots, M\}$. The sets are disjoint, so each resource function g_j is evaluated on at most one suffix.

For two requests arriving in the opposite order, a comparator first orders their prefixes, conditional swaps route the two suffixes to the prefix and suffix recovery branches, and a final swap restores the output order. This costs only $\text{poly}(n)$ gates.

The map $(f_0, \dots, f_{M-1}) \mapsto (g_0, \dots, g_M)$ is, up to an invertible change of information coordinates, the length- $(M+1)$ single-parity-check code. The two representations above are disjoint recovery sets for each information coordinate. This suggests replacing the one-dimensional telescoping code by a constant-rate code with many geometric recovery sets.

4 A constant-rate evaluation code

Fix an integer $\ell \geq 2$ and a field $\mathbb{F}_q$ of characteristic two, where $q = 2^b$. Set

$$s_0 = \left\lfloor \frac{q-1}{\ell} \right\rfloor$$

and fix a polynomial-basis representation of $\mathbb{F}_q$. If α_i denotes the field element whose canonical b -bit representation is the integer i , choose the efficiently indexed set

$$A = \{\alpha_0, \dots, \alpha_{s_0-1}\}.$$

Let $\mathcal{V}$ be the space of polynomials

$$P(X_1, \dots, X_\ell) \in \mathbb{F}_q[X_1, \dots, X_\ell]$$

having degree less than s_0 in every variable. Evaluation on A^ℓ uniquely determines P , and

$$\deg P \leq \ell(s_0 - 1) \leq q - 2.$$

The evaluation map

$$\text{Enc} : \mathbb{F}_q^{A^\ell} \longrightarrow \mathbb{F}_q^{\mathbb{F}_q^\ell}, \quad (P(u))_{u \in A^\ell} \longmapsto (P(y))_{y \in \mathbb{F}_q^\ell},$$

is a systematic linear code of length

$$N = q^\ell$$

and dimension

$$K = s_0^\ell = \Theta_\ell(q^\ell).$$

4.1 Packing Boolean restrictions

Split an n -bit input as

$$x = (a, z), \quad |a| = p, \quad |z| = d = n - p.$$

For each fixed suffix z , pack the 2^p bits

$$(f(a, z))_{a \in \{0,1\}^p}$$

into K field symbols, using the fixed $\mathbb{F}_2$ -basis of $\mathbb{F}_q$ and padding unused bit positions with zeros. More explicitly, interpret a prefix a as an integer $I(a) \in [2^p]_0$, and set

$$s(a) = \left\lfloor \frac{I(a)}{b} \right\rfloor, \quad j(a) = I(a) \bmod b.$$

Write $s(a)$ in base s_0 using exactly ℓ digits $i_1(a), \dots, i_\ell(a) \in [s_0]_0$, and define

$$u(a) = (\alpha_{i_1(a)}, \dots, \alpha_{i_\ell(a)}) \in A^\ell.$$

Thus bit $j(a)$ of the information symbol at $u(a)$ is $f(a, z)$. Fixed-divisor division, remainder, and base conversion have Boolean circuits of size $\text{poly}_\ell(p, \log q)$, so computing $(u(a), j(a))$ for all requests costs $\tilde{O}_\ell(t)$ gates.

Choose q , among powers of two, minimally so that

$$Kb \geq 2^p.$$

For all sufficiently large p (depending on ℓ), this also ensures $q > 2\ell + 2$, and minimality together with $K = \Theta_\ell(q^\ell)$ gives

$$q^\ell b = \Theta_\ell(2^p). \tag{1}$$

Indeed, the lower bound follows from $Kb \leq q^\ell b$. For the upper bound, the preceding candidate $q/2$ has bit width $b-1$, fails the packing inequality, and has dimension $\Omega_\ell(q^\ell)$. Hence $q^\ell(b-1) = O_\ell(2^p)$; since $b \leq 2(b-1)$ for $b \geq 2$, the upper bound in (1) follows. Equivalently,

$$\ell \log_2 q + \log_2 \log_2 q = p + O_\ell(1), \quad \log_2 q = \frac{p}{\ell} + O_\ell(\log p). \quad (2)$$

Let P_z be the polynomial whose information symbols are the packed bits. For each code coordinate $y \in \mathbb{F}_q^\ell$, define the field-valued resource

$$G_y(z) = P_z(y).$$

Each of the b coordinate bits of G_y is a Boolean function on d variables, determined by f , y , and the chosen field basis.

4.2 Affine-line recovery

Lemma 4.1 (Line identity). *Let $P : \mathbb{F}_q^\ell \rightarrow \mathbb{F}_q$ be represented by a polynomial of total degree at most $q-2$. For every $u \in \mathbb{F}_q^\ell$ and every nonzero direction $v \in \mathbb{F}_q^\ell$,*

$$P(u) = - \sum_{\lambda \in \mathbb{F}_q^*} P(u + \lambda v).$$

In characteristic two, the minus sign may be omitted.

Proof. The restriction $Q(\lambda) = P(u + \lambda v)$ is a univariate polynomial of degree at most $q-2$. The sum of the constant monomial over $\mathbb{F}_q$ is $q \cdot 1 = 0$ in characteristic two. For $1 \leq j \leq q-2$, the sum of λ^j over $\mathbb{F}_q^*$ vanishes because j is not divisible by $q-1$, and the zero term contributes nothing. Applying these identities monomial by monomial gives $\sum_{\lambda \in \mathbb{F}_q} Q(\lambda) = 0$. Separating the term $\lambda = 0$ proves the claim. $\square$

Thus every affine line through an information point u supplies a recovery set consisting of its $q-1$ non-target points. The number of projective directions through a point is

$$D_q = \frac{q^\ell - 1}{q - 1} = 1 + q + \cdots + q^{\ell-1}, \quad q^{\ell-1} \leq D_q < 2q^{\ell-1}. \quad (3)$$

For fixed ℓ and $p \rightarrow \infty$, Equations (2) and (3) give the parameter dictionary

$$q^\ell = 2^{p+o(p)}, \quad q = 2^{p/\ell+o(p)}, \quad D_q = q^{\ell-1+o(1)} = 2^{((\ell-1)/\ell)p+o(p)}. \quad (4)$$

Two distinct lines through the same target intersect only at that target, which is absent from both recovery sets. Repeated requests for one information symbol therefore cause no additional difficulty.

5 Deterministic scheduling and routing

5.1 Disjoint scheduling inside one group

Consider a multiset of g requested information points $u_1, \dots, u_g \in A^\ell$. Process the requests in order. Suppose disjoint recovery sets have been selected for requests $1, \dots, j-1$, and let U be their union. Then

$$|U| = (j-1)(q-1) < gq.$$

For the target u_j , every point $y \in U \setminus \{u_j\}$ forbids at most one projective direction, namely the direction of $y - u_j$. The point u_j itself forbids none because it is omitted from the new recovery set.

Lemma 5.1 (Projective indexing). *Projective directions in $\mathbb{F}_q^\ell$ have rank and unrank circuits of size $\text{poly}_\ell(b)$, with ranks in $[D_q]_0$.*

Proof. Normalize a nonzero vector so that its first nonzero coordinate is one. The normalized directions whose first nonzero coordinate is in position $h \in \{1, \dots, \ell\}$ have the form

$$(0, \dots, 0, 1, w_{h+1}, \dots, w_\ell)$$

and form a block of size $q^{\ell-h}$. Let

$$B_h = \sum_{r=1}^{h-1} q^{\ell-r}, \quad \text{val}_q(w_{h+1}, \dots, w_\ell) = \sum_{r=h+1}^{\ell} \text{int}(w_r) q^{\ell-r},$$

where $\text{int}(w_r) \in [q]_0$ is the canonical binary representation. The rank is

$$B_h + \text{val}_q(w_{h+1}, \dots, w_\ell).$$

The blocks are consecutive and their total size is D_q .

To rank, find the first nonzero coordinate, invert it, normalize the vector, and concatenate the b -bit tail coordinates. To unrank, compare the rank with the ℓ hardwired block intervals, subtract the appropriate B_h , and split the remainder into base- q digits. Since $q = 2^b$, the last step is bit slicing. The field operations and comparisons from the circuit conventions give size $\text{poly}_\ell(b)$ in both directions. $\square$

Lemma 5.2 (Greedy line scheduler). *If*

$$gq < D_q, \tag{5}$$

then every multiset of g target points admits pairwise disjoint affine-line recovery sets. Given the targets, the recovery sets can be selected by a Boolean circuit of size

$$\tilde{O}_\ell(g^2 q).$$

Proof. At step j , fewer than D_q directions are forbidden, so a valid direction exists. This proves the combinatorial statement.

For the circuit bound, represent a projective direction by normalizing its first nonzero coordinate to one and use the ranking from [Lemma 5.1](#). At stage j , for each $y \in U$, output the sentinel rank D_q when $y = u_j$ and otherwise output the projective rank of $y - u_j$. At the gate level, first multiplex a fixed nonzero vector in place of $y - u_j$ when $y = u_j$, rank that always-nonzero vector, and then overwrite the result by D_q on the equality branch. Use $\lceil \log_2(D_q + 1) \rceil$ bits per rank. There are $(j-1)(q-1) = O(jq)$ such entries.

Sort these ranks as $r_1 \leq \dots \leq r_m$. The least missing rank can be found without a dense D_q -bit table. Rank zero is a candidate if $m = 0$ or $r_1 > 0$; after position i , rank $r_i + 1$ is a candidate if

$$r_i < D_q - 1 \quad \text{and} \quad (i = m \text{ or } r_{i+1} > r_i + 1).$$

These tests are applied to the full sorted list, including duplicates and sentinels. Only the last occurrence of a non-sentinel rank can generate its successor, and a sentinel cannot generate a candidate. A priority circuit selects the first candidate in this order. A left-to-right prefix OR and multiplexers implement the choice in $O(m \log D_q)$ gates. Since $|U \setminus \{u_j\}| < D_q$, fewer than D_q distinct non-sentinel ranks are forbidden, so a candidate exists. Sorting, comparisons, and priority selection therefore cost $\tilde{O}_\ell(jq)$ gates. Unrank the selected direction using [Lemma 5.1](#) and enumerate its $q - 1$ non-target line points, at cost $q \text{poly}_\ell(b)$. Unrolling the stages and summing over $j \leq g$ gives $\tilde{O}_\ell(g^2q)$. $\square$

5.2 Request groups

Partition the t requests into C fixed groups, each of size at most

$$g = \left\lceil \frac{t}{C} \right\rceil.$$

Apply [Lemma 5.2](#) independently inside each group. A code coordinate is used at most once per group and therefore at most C times overall. The total scheduling cost is

$$C \cdot \tilde{O}_\ell(g^2q) = \tilde{O}_\ell\left(\frac{t^2q}{C} + Cq\right), \quad (6)$$

because

$$C \left\lceil \frac{t}{C} \right\rceil^2 q \leq \frac{t^2q}{C} + 2tq + Cq \leq 2 \left(\frac{t^2q}{C} + Cq \right).$$

The last inequality is the arithmetic–geometric mean inequality. The construction is valid whenever (5) holds for $g = \lceil t/C \rceil$, that is, whenever

$$\left\lceil \frac{t}{C} \right\rceil q < D_q. \quad (7)$$

5.3 Scatter and gather

Every non-target point of a selected line produces an incidence record containing a request identifier, a group, a line position, the point’s code coordinate y , and the suffix to be evaluated. There are exactly $t(q - 1) < tq$ such records. Inside one group, each coordinate appears in at most one record.

The routing consists of four fixed sorting-and-local-update passes. Here “key” means (group, y); any unlisted tie is broken by the record’s original position.

Pass	Records	Sort order	Local result
Scatter match	incidence, slot	(key, type); incidence before slot	a matched slot takes its predecessor’s suffix; otherwise it takes a dummy
Scatter order	incidence, filled slot	(type, key); slots first	slots are in canonical (group, y) order
Gather match	value, saved incidence	(key, type); value before incidence	a matched incidence takes its predecessor’s value
Gather order	value, updated incidence	(type, request, line position); incidences first	decoder inputs are in canonical request order

Scatter records carry a d -bit suffix. After resource evaluation, value records carry the b -bit field value $G_y(z)$. The gather passes are payload-agnostic for any payload whose width is polynomial in the displayed parameters.

Lemma 5.3 (Record routing). *The suffixes can be scattered into slots indexed by $(\text{group}, y) \in [C]_0 \times \mathbb{F}_q^\ell$, and the resulting resource values can be gathered back to their request records, using circuits of total size*

$$\tilde{O}_\ell(tq + Cq^\ell).$$

Proof. For scatter, create one slot record for each of the Cq^ℓ keys and combine them with the fewer than tq incidence records. Preserve a copy of every incidence record by free fan-out. Uniqueness inside a group puts at most one incidence at each key. After the first pass in the table, a slot therefore copies the preceding suffix exactly when a type-and-key equality test succeeds; otherwise it receives a fixed dummy suffix. The predecessor of the first position is defined to be a dummy record, so the boundary case uses the same guard. The second pass places the filled slots in canonical (group, y) order. Fixed wiring transposes this order to (y, group) , the input order used by the resource circuits.

For gather, combine the preserved incidences with one value record per key. The third pass places that value immediately before every matching incidence; a guarded local multiplexer copies it into the incidence. The fourth pass puts the first $t(q-1)$ positions in the fixed request-and-line order required by the decoders. Thus absent incidences, the first array position, and all type boundaries are handled by fixed dummy data and the same guarded update, not by data-dependent wiring.

This is a constant number of sorting networks on $O(tq + Cq^\ell)$ records. Each record has width $\text{poly}_\ell(n, \log q, \log t, \log C)$, so the standard primitives from the circuit conventions give the stated bound. $\square$

The lemma is the reason no tq^ℓ crossbar appears. Only the actual incidences and one dummy record per resource slot are routed.

6 The master recurrence

Bounded-demand invariant. Fix f and the packing above, and for $y \in \mathbb{F}_q^\ell$ and $k \in [b]_0$ let

$$H_{y,k}(z) = (G_y(z))_k.$$

This is one fixed Boolean function on d variables. Inside each request group, disjoint recovery ensures that coordinate y occurs at most once; hence the pair (y, k) receives at most one suffix from each of the C groups. Place those suffixes in the canonical group slots and pad unused slots with a fixed dummy suffix. One circuit for $H_{y,k}^{\times C}$, of size at most $L_C(d)$, serves exactly this bounded demand. There are $q^\ell b$ such pairs, and no sharing between different pairs is assumed.

Proposition 6.1 (Master bound). *Fix $\ell \geq 2$. Let $n = p + d$ with p sufficiently large depending on ℓ , choose $q = 2^b$ minimally so that $s_0^\ell b \geq 2^p$, and suppose [Equation \(7\)](#) holds. Then every $f : \{0, 1\}^n \rightarrow \{0, 1\}$ satisfies*

$$C(f^{\times t}) \leq q^\ell b L_C(d) + \tilde{O}_\ell\left(\frac{t^2 q}{C} + tq + Cq^\ell\right). \quad (8)$$

Proof. Map every requested prefix a to its information point $u(a)$ and bit position $j(a)$ using the canonical packing circuit. In each request group, use [Lemma 5.2](#) to select and enumerate a recovery line. Together these steps cost $\tilde{O}_\ell(t) + \tilde{O}_\ell(t^2q/C + Cq)$ gates. Each non-target point y on a selected line requests G_y on the suffix of the corresponding output instance.

For each pair (y, k) , use the $H_{y,k}^{\times C}$ circuit from the bounded-demand invariant on the canonical group slots. The definition of $L_C(d)$ applies separately to each induced function $H_{y,k}$, so these circuits cost $q^\ell b L_C(d)$ in total. The complete scatter, gather, and fixed-wire compaction pipeline costs $\tilde{O}_\ell(tq + Cq^\ell)$ by [Lemma 5.3](#). Field addition is coordinatewise XOR in the fixed $\mathbb{F}_2$ basis, so XORing the $q - 1$ returned b -bit values of a request and selecting bit $j(a)$ costs $O(qb)$ per request and $O(tqb)$ in total. Since each selected line passes through its target $u(a)$ and $\deg P_z \leq q - 2$, [Lemma 4.1](#) shows that this XOR equals $P_z(u(a))$ for the request's suffix z , and bit $j(a)$ of that symbol is $f(a, z)$ by the packing of [Section 4.1](#). The circuit therefore computes $f^{\times t}$.

The polynomial P_z and the functions G_y specify the nonuniform resource truth tables; no run-time encoder is being assumed. Since $q \geq 2$, all smaller terms in this gate ledger are absorbed by the overhead in (8), proving the bound. $\square$

7 The exponent bootstrap

Definition 7.1 (Admissible exponent property). For $0 < \beta \leq 1$, let $\text{Adm}(\beta)$ denote the following assertion. For every fixed $0 < \rho < \beta$, there is a constant $A_{\beta,\rho}$ such that, for all sufficiently large m , every $h : \{0, 1\}^m \rightarrow \{0, 1\}$, and every integer $1 \leq r \leq 2^{\rho m}$,

$$C(h^{\times r}) \leq A_{\beta,\rho} \frac{2^m}{m}.$$

The constant and the threshold may depend on the fixed exponents β and ρ , but not on m , h , or r .

Lemma 7.2 (Strict exponential margins). Suppose $X_n, Y_n > 0$ and, for fixed $a < b$,

$$\log_2 X_n \leq an + o(n), \quad \log_2 Y_n \geq bn + o(n).$$

Then $R(n)X_n < Y_n$ for every fixed polynomial R and all sufficiently large n . Also, for every fixed $a \geq 0$ and $\eta > 0$, $\lceil 2^{an} \rceil \leq 2^{(a+\eta)n}$ for all sufficiently large n .

Proof. Choose a positive margin smaller than $(b - a)/3$. The two $o(n)$ terms and $\log_2 R(n) = o(n)$ are eventually smaller than this margin times n , which proves the first claim. The second follows from $\lceil 2^{an} \rceil \leq 2^{an} + 1 \leq 2^{an+1}$ and $an + 1 \leq (a + \eta)n$ once $n \geq 1/\eta$. $\square$

7.1 The disjoint base range

Lemma 7.3 (Base range). $\text{Adm}(1/2)$ holds.

Proof. Fix $0 < \gamma < 1/2$ and let $t \leq 2^{\gamma n}$. Choose

$$d = \lfloor \gamma n \rfloor, \quad p = n - d, \quad C = 1,$$

and write $\vartheta = 1 - \gamma$. By Equation (4),

$$\log_2 q = \frac{\vartheta n}{\ell} + o(n), \quad \log_2 q^\ell = \vartheta n + o(n).$$

Choose a sufficiently large fixed ℓ so that

$$\gamma + \frac{\vartheta}{\ell} < \vartheta \left(1 - \frac{1}{\ell}\right). \quad (9)$$

This is possible precisely because $\gamma < 1/2$. Equations (3) and (9), together with Lemma 7.2, imply $tq < D_q$, so the scheduler is valid for all sufficiently large n .

The resource term in Equation (8) is, by Shannon–Lupanov synthesis,

$$q^\ell bL(d) = O_\ell \left(2^p \frac{2^d}{d} \right) = O_\gamma(2^n/n).$$

The scheduler exponent is

$$2\gamma + \frac{\vartheta}{\ell} < 1,$$

where the strict inequality follows from the stronger margin (9). The tq and q^ℓ terms also have exponents below one. After the suppressed polynomial factors are restored, every overhead term is $2^{(1-\Omega_\gamma(1))n}$ and is negligible compared with $2^n/n$ by Lemma 7.2. The same strict margins absorb the floor in d and the field-size rounding. $\square$

7.2 Recursive congestion reduction

Lemma 7.4 (Bootstrap). *If $\text{Adm}(\beta)$ holds for some $0 < \beta \leq 1$, then*

$$\text{Adm}\left(\frac{1}{2-\beta}\right)$$

holds.

Proof. To verify the asserted property, fix a target exponent $0 < \gamma < 1/(2-\beta)$. The range $\gamma < 1/2$ is already covered by Lemma 7.3. If $\gamma = 1/2$, choose $\hat{\gamma} \in (1/2, 1/(2-\beta))$; the construction below for $\hat{\gamma}$ is uniform for every $t \leq 2^{\hat{\gamma}n}$ and hence, by monotonicity of the allowed range, also handles every $t \leq 2^{\gamma n}$. We may therefore assume

$$\frac{1}{2} < \gamma < \frac{1}{2-\beta}.$$

The upper inequality is equivalent to

$$2\gamma - 1 < \beta\gamma.$$

Choose a constant c such that

$$2\gamma - 1 < c < \beta\gamma. \quad (10)$$

Put

$$d = \lfloor \gamma n \rfloor, \quad p = n - d, \quad C = \lceil 2^{cn} \rceil.$$

Choose ρ with

$$\frac{c}{\gamma} < \rho < \beta.$$

Since $c < \rho\gamma$, the bounds $C \leq 2^{cn} + 1 \leq 2^{cn+1}$ and $d \geq \gamma n - 1$ imply $C \leq 2^{\rho d}$ for all sufficiently large n . Thus $\text{Adm}(\beta)$ gives

$$L_C(d) = O_{\beta,\rho}(2^d/d).$$

By Equation (1), the resource term is therefore

$$q^\ell b L_C(d) = O_{\beta,\gamma,\rho,\ell}(2^n/n).$$

Set $\vartheta = 1 - \gamma$. The request-group size has exponent $\gamma - c$. Choose a sufficiently large fixed ℓ so that

$$c > 2\gamma - 1 + \frac{2\vartheta}{\ell}. \quad (11)$$

This is possible by the left inequality in (10). The scheduling condition follows because

$$\gamma - c + \frac{\vartheta}{\ell} < \vartheta \left(1 - \frac{1}{\ell}\right).$$

Indeed, $\lceil t/C \rceil \leq 2^{(\gamma-c)n} + 1$, while Equations (3) and (4) and Lemma 7.2 turn this strict exponent margin into Equation (7) for all sufficiently large n .

The three overhead exponents in Equation (8) are bounded by

$$\begin{aligned} \frac{1}{n} \log_2 \left(\frac{t^2 q}{C} \right) &\leq 2\gamma - c + \frac{\vartheta}{\ell} + o(1) < 1, \\ \frac{1}{n} \log_2(tq) &\leq \gamma + \frac{\vartheta}{\ell} + o(1) < 1, \\ \frac{1}{n} \log_2(Cq^\ell) &= c + \vartheta + o(1) < 1. \end{aligned}$$

The first inequality follows from (11), the second from $\ell > 1$, and the third from $c < \beta\gamma \leq \gamma$. Thus every overhead term is exponentially smaller than $2^n/n$ by Lemma 7.2. All choices in this one-step argument are fixed constants. In the final finite iteration they depend only on the target exponent γ , so their constants combine into A_γ . $\square$

Proof of Theorem 1.1. The case $\gamma = 0$ is the Shannon–Lupanov upper bound. Starting from $\beta_1 = 1/2$, define

$$\beta_{k+1} = \frac{1}{2 - \beta_k}.$$

An induction gives $\beta_k = k/(k+1)$. By Lemmas 7.3 and 7.4, $\text{Adm}(\beta_k)$ holds for every k . Given fixed $0 < \gamma < 1$, choose k such that $\gamma < k/(k+1)$ and apply $\text{Adm}(\beta_k)$ with $\rho = \gamma$. $\square$

Remark 7.5 (What is gained and what is lost). Uhlig’s theorem permits $2^{o(n/\log n)}$ copies while preserving the basis-dependent one-copy leading asymptotic. Theorem 1.1 permits $2^{(1-\varepsilon)n}$ copies for every fixed $\varepsilon > 0$, but only with a constant $A_{1-\varepsilon}$ for which no control is claimed as $\varepsilon \rightarrow 0$. Relative to that benchmark, the tradeoff is a larger exponent for the number of copies without control of the leading circuit-size constant.

8 A barrier to copy-superadditive measures

Corollary 8.1 (Copy-superadditive barrier). *Let μ be a nonnegative complexity measure on Boolean vector functions such that*

$$\mu(F) \leq C(F)$$

for every F , and

$$\mu(f^{\times r}) \geq r\mu(f)$$

for every Boolean function f and every $r \geq 1$. Then, for every fixed $\varepsilon > 0$ and every $f : \{0, 1\}^n \rightarrow \{0, 1\}$,

$$\mu(f) = O_\varepsilon(2^{\varepsilon n}/n).$$

Equivalently, the worst-case value is uniformly subexponential:

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log_2 \left(1 + \max_{f: \{0,1\}^n \rightarrow \{0,1\}} \mu(f) \right) = 0.$$

Proof. It is enough to consider $0 < \varepsilon < 1$, since for $\varepsilon \geq 1$ the claim follows directly from $\mu(f) \leq C(f) = O(2^n/n)$. For $0 < \varepsilon < 1$, apply [Theorem 1.1](#) with $r = \lfloor 2^{(1-\varepsilon)n} \rfloor$. Then

$$r\mu(f) \leq \mu(f^{\times r}) \leq C(f^{\times r}) \leq A_{1-\varepsilon} \frac{2^n}{n}.$$

For all sufficiently large n , $r \geq 2^{(1-\varepsilon)n-1}$, so dividing by r proves the first claim; the finitely many smaller values of n are absorbed into the constant. Because the bound is uniform in f and holds for every fixed $\varepsilon > 0$, the second claim follows. $\square$

For comparison, applying Paul's bound with $r = \lfloor 2^{(1-\varepsilon)n} \rfloor$ gives

$$\mu(f) = O_\varepsilon \left(\frac{n2^n}{r} + n \log^2 r \right) = O_\varepsilon(n2^{\varepsilon n} + n^3) = O_\varepsilon(n2^{\varepsilon n}).$$

Thus the qualitative subexponential barrier predates [Theorem 1.1](#); the present theorem sharpens its polynomial prefactor by a factor of n^2 .

The corollary does not obstruct lower-bound measures that fail this copy-superadditivity condition or that explicitly track the possibility of cross-instance sharing. Rather, it identifies a structural requirement: a successful unrestricted-circuit measure cannot assign an independent, nonshareable charge to every copy of a function.

9 Discussion and limitations

Historical comparison. The introduction gives theorem-level comparisons with the verified Uhlig and Paul bounds and records the limited comparison that the available primary-source evidence supports for Albrecht. Because Albrecht's quantitative theorems have not been checked from the full text, this manuscript deliberately makes no historical priority claim. This limitation concerns the novelty comparison, not the self-contained argument proved here.

Leading constants. The resource term pays for q^ℓ code symbols while carrying s_0^ℓ information symbols. Since s_0/q tends to $1/\ell$ for fixed ℓ as q grows, the ratio q^ℓ/s_0^ℓ is roughly ℓ^ℓ . Moreover, synthesizing the resource functions separately introduces the factor n/d , which is approximately $1/\gamma$ in a bootstrap step. Thus, normalized by $2^n/n$, the recursive coefficient is multiplied by roughly ℓ^ℓ/γ even before field-size rounding, and repeated bootstrapping compounds the loss. The scheduler and routing terms are exponentially lower order for every fixed exponent margin, so improving their constant factors alone would not recover a sharp leading term. Our argument therefore does not determine whether the one-copy Shannon–Lupanov leading constant for this basis can be retained for any fixed exponential number of copies. We make no claim about whether the existing literature settles that question.

The endpoint. The theorem reaches $2^{(1-\varepsilon)n}$ copies for every fixed $\varepsilon > 0$, but does not treat a rate that approaches one with n . Some degradation in that regime is forced. If $n \geq 2$ and f depends on all n of its variables, every output of a circuit for $f^{\times t}$ is a gate, and each of the tn inputs must feed at least one gate, since otherwise the corresponding output would not depend on it. Since gates have fan-in at most two,

$$C(f^{\times t}) \geq \frac{tn}{2}.$$

A bound of the form $A 2^n/n$ therefore cannot hold once $t > 2A \cdot 2^n/n^2$; in particular, a uniform $O(2^n/n)$ plateau must fail for $t = \omega(2^n/n^2)$. The theorem still leaves the detailed mass-production curve throughout the near-endpoint regime $t = 2^{n-o(n)}$ unresolved. Determining how far the plateau extends, and then characterizing the growth beyond it, would require a quantitatively sharper schedule or a different recursive decomposition.

Restricted models. The construction uses arbitrary routing, finite-field linear reconstruction, and free fan-out. Formula size, bounded fan-out circuits, monotone circuits, bounded-depth circuits, and branching programs may have different mass-production curves. In those settings, simultaneous demand may become a source of genuine direct-sum lower bounds.

Other recovery geometries. Affine lines provide many recovery choices with simple intersection behavior, but their recovery sets have size $q - 1$. Batch codes, locally recoverable codes, or higher-dimensional varieties may improve the group-size versus decoding-cost tradeoff and reduce the number of bootstrap rounds.

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